HIDDEN SIGNATURES: USING CROSSCODERS TO COMPARE TRANSFORMER FEATURES ACROSS MODELS

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Problem & Approach

Challenge: Understanding how model compression affects internal computational mechanisms

Solution: Crosscoder-based feature diffing across transformation types to identify shared vs exclusive features

Impact: Reveals compression mechanisms and safety implications

How Crosscoders Work

Crosscoders learn to map features from **Model A → Shared Space → Model B**. They read from and write to multiple layers/models simultaneously.

Training Objective:
Reconstruct activations from both models simultaneously, creating a common feature vocabulary for direct comparison. Architecture: 512→1024→512, trained for 10k steps

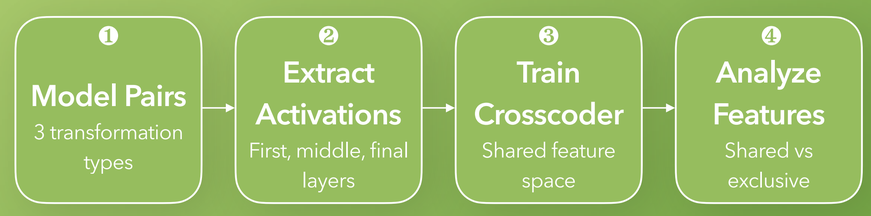
Methods

Models: Gemma-2-2B (base + instruct), GPT-2 (teacher + distilled student), Pythia (160M + 70M)

Layers: 0, 12, 24 for Gemma; 0, 6, 12 for GPT-2; 0, 6, 12 for Pythia

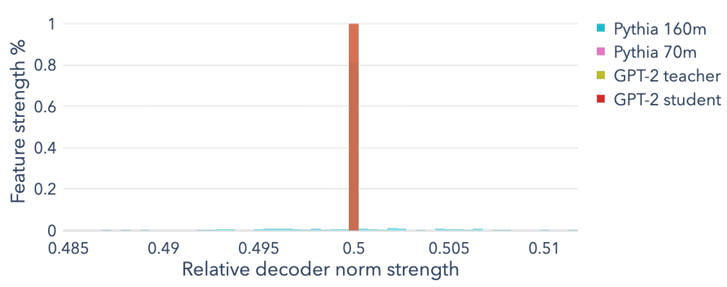
Crosscoders: 512→1024→512 architecture, 10k training steps
Data: Same activation samples across all model pairs for fair comparison

Model Diffing pipeline

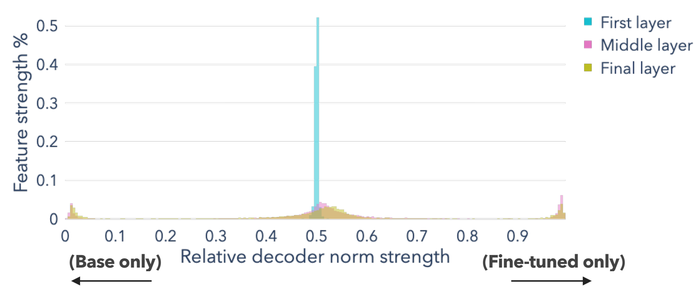


Results

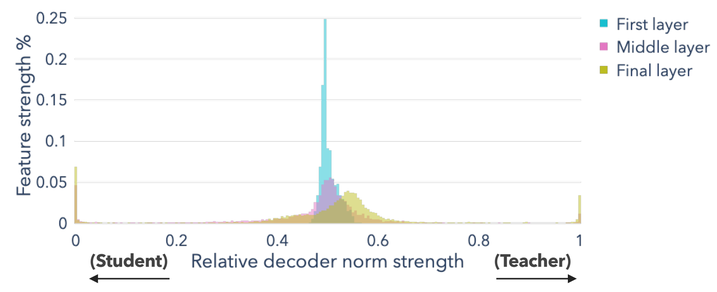
Same Model Comparison: Decoder Activation Norms



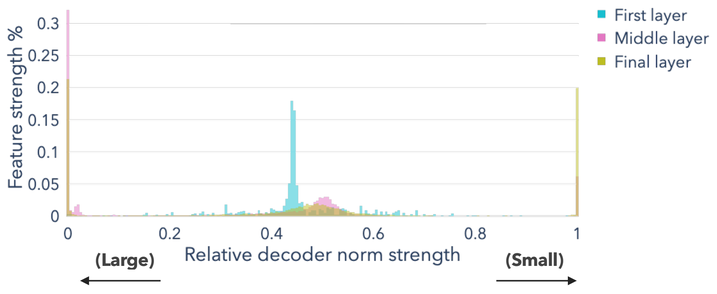
Gemma Fine-tuning: Decoder Activation Norms



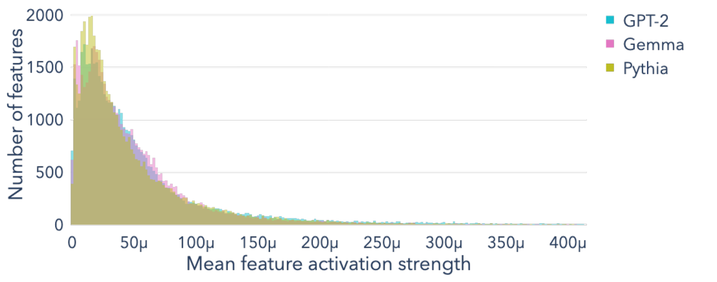
GPT-2 Distillation: Decoder Activation Norms



Pythia Model Sizes: Decoder Activation Norms



Dataset Feature Activation Density



BCE: BCE measures to what degree features are shared versus model specific.
CCA: Measures representational similarity (how aligned feature spaces are).
Interpretation: BCE high, CCA low = good reconstruction but different organization.

Key Findings

❶ Fine-tuning Pattern
Early layers show 99.9% similarity
CCA: 0.999 → 0.807. Early layers extract universal features, late layers specialize for specific tasks.

❷ Distillation Anomaly
U-shaped similarity pattern
CCA: 0.937 → 0.604 → 0.715. Middle layers most affected by compression process.

❸ Scale Baseline
Low similarity (28-35%) confirms method
CCA: 0.286 → 0.353. Different architectures properly detected as dissimilar.

Similarity Scores BCE/ CCA

Transformation	First Layer	Middle Layer	Final layer
Gemma Fine-tuning	1.00 / 0.999	0.77 / 0.768	0.76 / 0.807
GPT-2 Distillation	1.00 / 0.937	0.91 / 0.604	0.85 / 0.715
Pythia Scale	0.96 / 0.286	0.54 / 0.343	0.54 / 0.353
Sanity Check	1.00 / 1.000	1.00 / 1.000	1.00 / 1.000

Why CCA Works Better

Canonical Correlation Analysis provides robust similarity metrics across different model sizes, unlike cosine similarity.
BCE-CCA Disagreement: measures to what degree features are shared versus model specific., CCA measures representational alignment. Distillation middle layer: good reconstruction (BCE=0.91) but reorganized features (CCA=0.60).

References: [1] Lindsey et al. Sparse Crosscoders for Cross-Layer Features and Model Diffing. Transformer Circuits Thread, 2024. • [2] Ojha et al. What Knowledge Gets Distilled in Knowledge Distillation? arXiv:2205.16004, 2023.